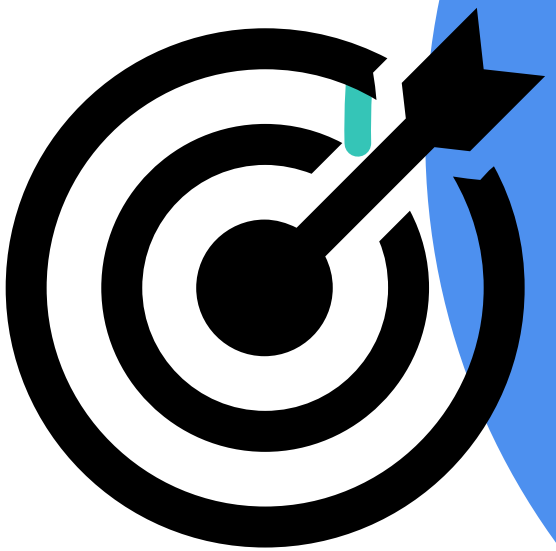
Machine Learning Systems Design : Overview



เป้าหมาย :
กระตุ้น ดลใจ
ML System

Agenda

- Why ML is so Popular
- ML Research vs. ML Production
- Intro to ML Dev
- ML Systems Design
- ML Systems vs Traditional Software
- How ML works
- ML Systems Design

Why ML is so Popular

Am I left behind if I have no ML?

ตกรถไฟขบวนสุดท้ายไหม?

นับจากปี 2016 … บางสิ่งบางอย่างได้เกิดขึ้น

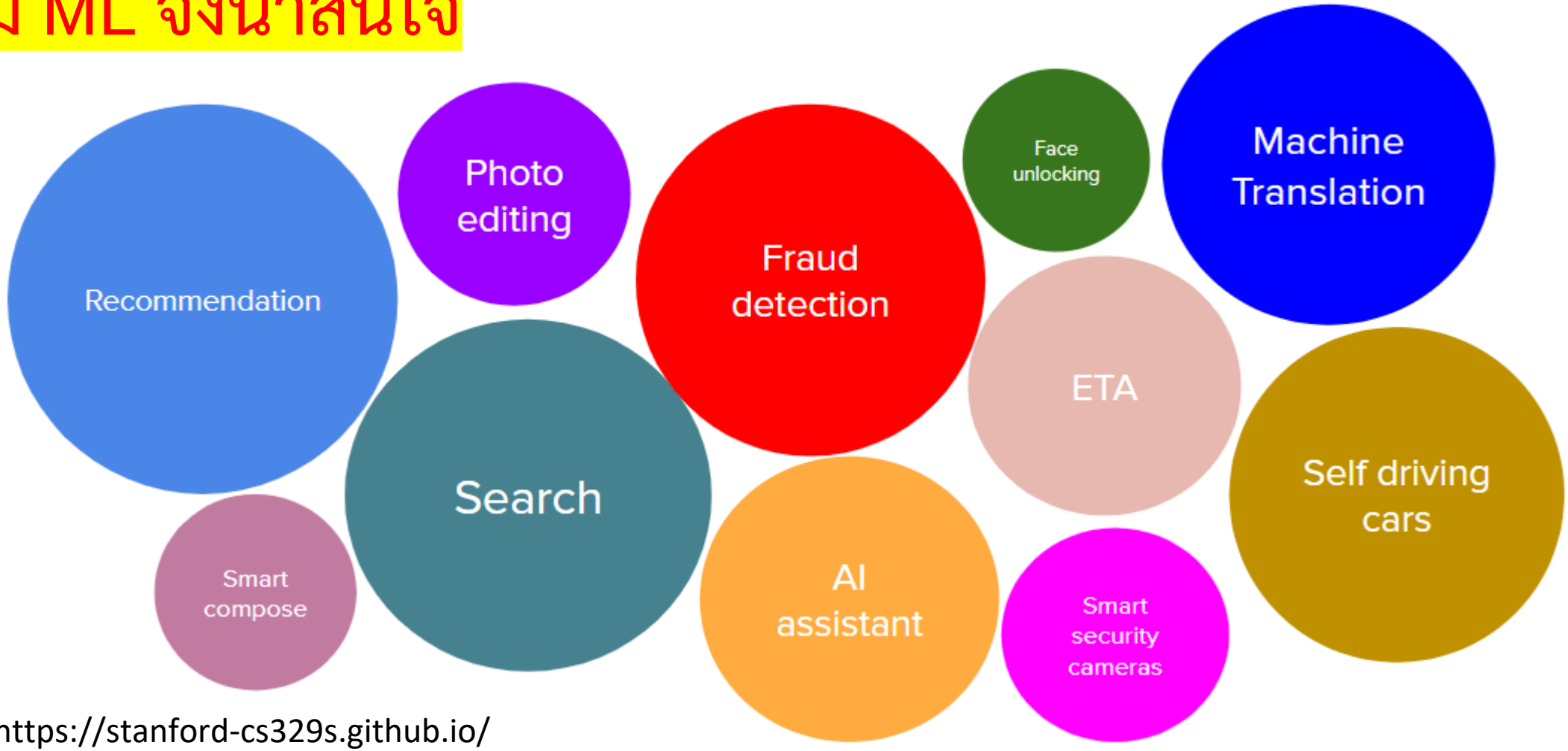
- In November 2016, Google announced that it had incorporated its multilingual neural machine translation system into Google Translate, marking one of the first success stories of deep neural artificial neural networks in production at scale
- According to Google, with this update, Google Translate ’s quality of translation improved more in a single leap than they had seen in the previous ten years combined
- In just five years (2022), ML has found its way into almost every aspect of our lives, from how we access information,
how we communicate, how we work, to how we find love. The spread of ML has been so rapid that it’s already hard to imagine life without it.

2022 : ปีทองแห่ง ML

https://stanford-cs329s.github.io/

2022: ML is in almost every aspect of our lives

ทำไม ML จึงน่าสนใจ

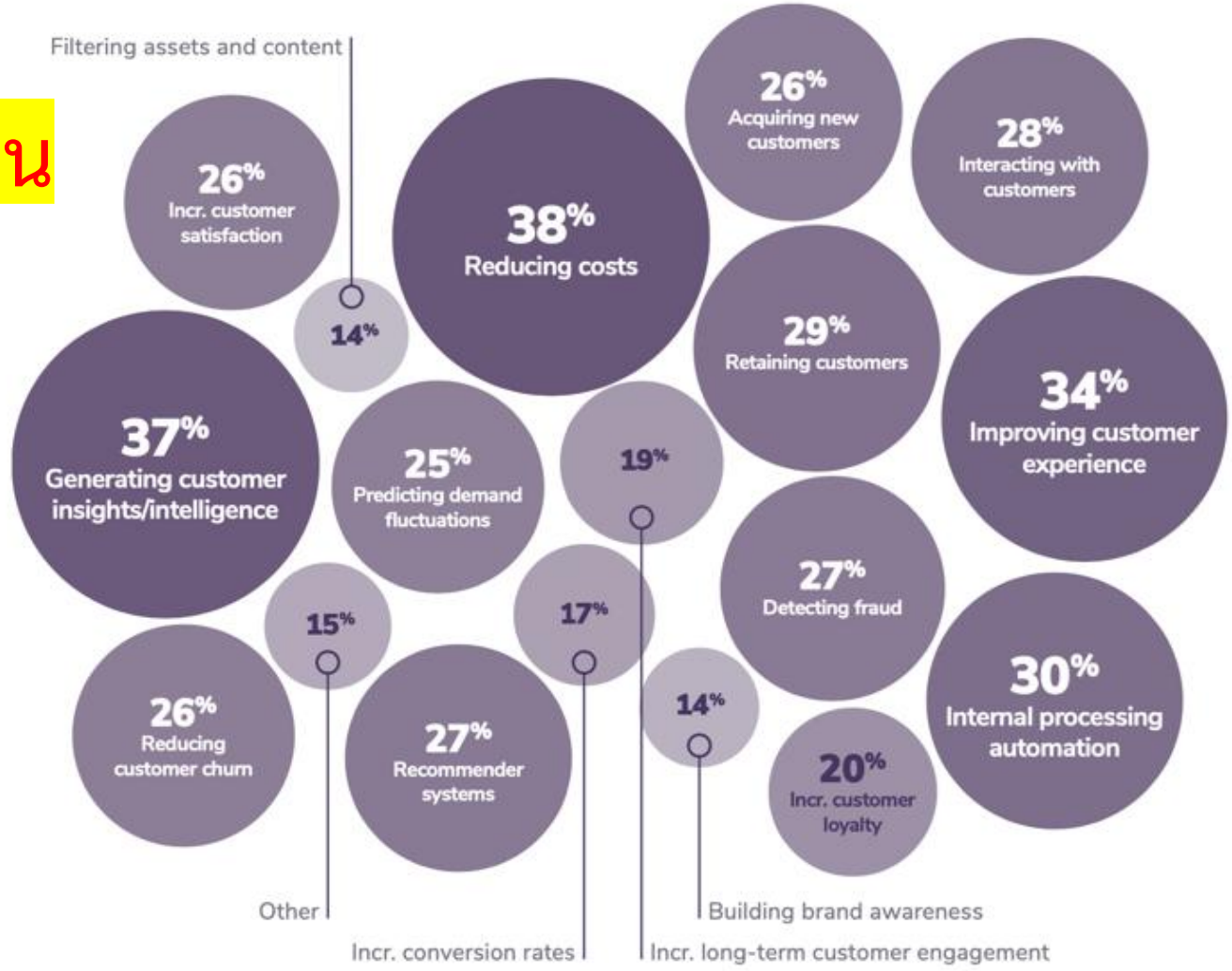


https://stanford-cs329s.github.io/

Enterprise use cases

เทียบเป็น % การใช้งาน
แต่ละ use case

Machine learning use case frequency



See course website: https://cs329s.stanford.edu

(ตัวอย่าง) 2022...ปีทองแห่ง Generative AI

เครื่องมือ	ผู้พัฒนา	รูปแบบข้อมูลที่สร้าง	ปีที่เปิดตัวเด่น
ChatGPT	OpenAI	ข้อความ, โต้ตอบ	พ.ย. 2022 (เปิดตัวสู่สาธารณะ)
Gemini (เดิมคือ Bard)	Google	ข้อความ, โต้ตอบ (Multimodal)	มี.ค. 2023
Claude	Anthropic	ข้อความ, โต้ตอบ	มี.ค. 2023
LLaMA	Meta	โมเดลภาษา Open-source	2023

https://stanford-cs329s.github.io/

ดู VDO
AI Trends in 2022
(ถ้ามีเวลา)

https://www.youtube.com/watch?v=ncWurMlYH6U

What is ML?

Machine Learning is

มี keywords อยู่ 5 ตัว

Machine learning is an approach to
(1) learn
(2) complex
(3) patterns from
(4) existing data and use these patterns to make
(5) predictions on
(6) unseen data.

ML research vs. ML production

ML ในห้องทดลองใช้งานธุรกิจได้จริงไหม

ML research vs. ML production

	Research	**Production**
Objectives	Model performance*	Different stakeholders have different objectives

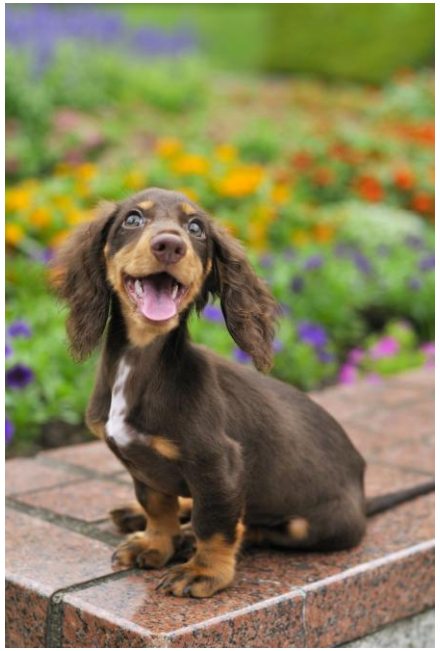
เทียบด้านเป้าประสงค์จากผู้ใช้จริง

"*" It's actively being worked. See Utility is in the Eye of the User: A Critique of NLP Leaderboards (Ethayarajh and Jurafsky, EMNLP 2020)

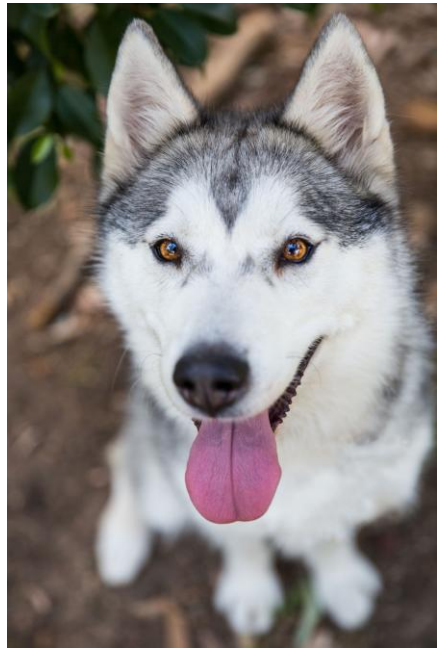
Stakeholder objectives

ผู้ใช้จริงมีส่วนได้ส่วนเสีย
ต่างมีเป้าประสงค์ต่างกัน

ML team
highest accuracy

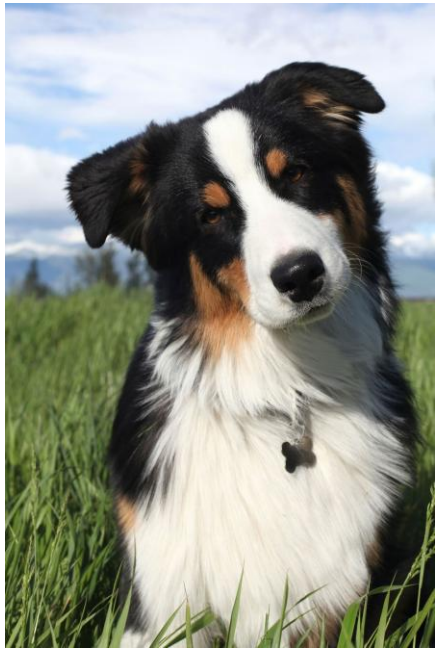


Sales
sells more ads



Product
fastest inference

Manager
maximizes profit = laying off ML teams



Leaderboard-style ML

สิ่งที่ต้องเพิ่มเติม
คาดหวังจากผู้ใช้จริง

- More comprehensive utility function
  - Model performance (e.g. accuracy)
  - Latency
  - Prediction cost
  - Interpretability
  - Robustness
  - Ease of use (e.g. OSS tools, community support)
  - Hardware requirements
- Adaptive to different use cases
  - Instead of a leaderboard for each dataset/task, the leaderboard adapts to each company's needs
- Dynamic datasets
  - Realistic distribution shifts with different types of shifts

Computational priority

	Research	**Production**
Objectives	Model performance	Different stakeholders have different objectives
Computational priority	Fast training, high throughput	Fast inference, low latency

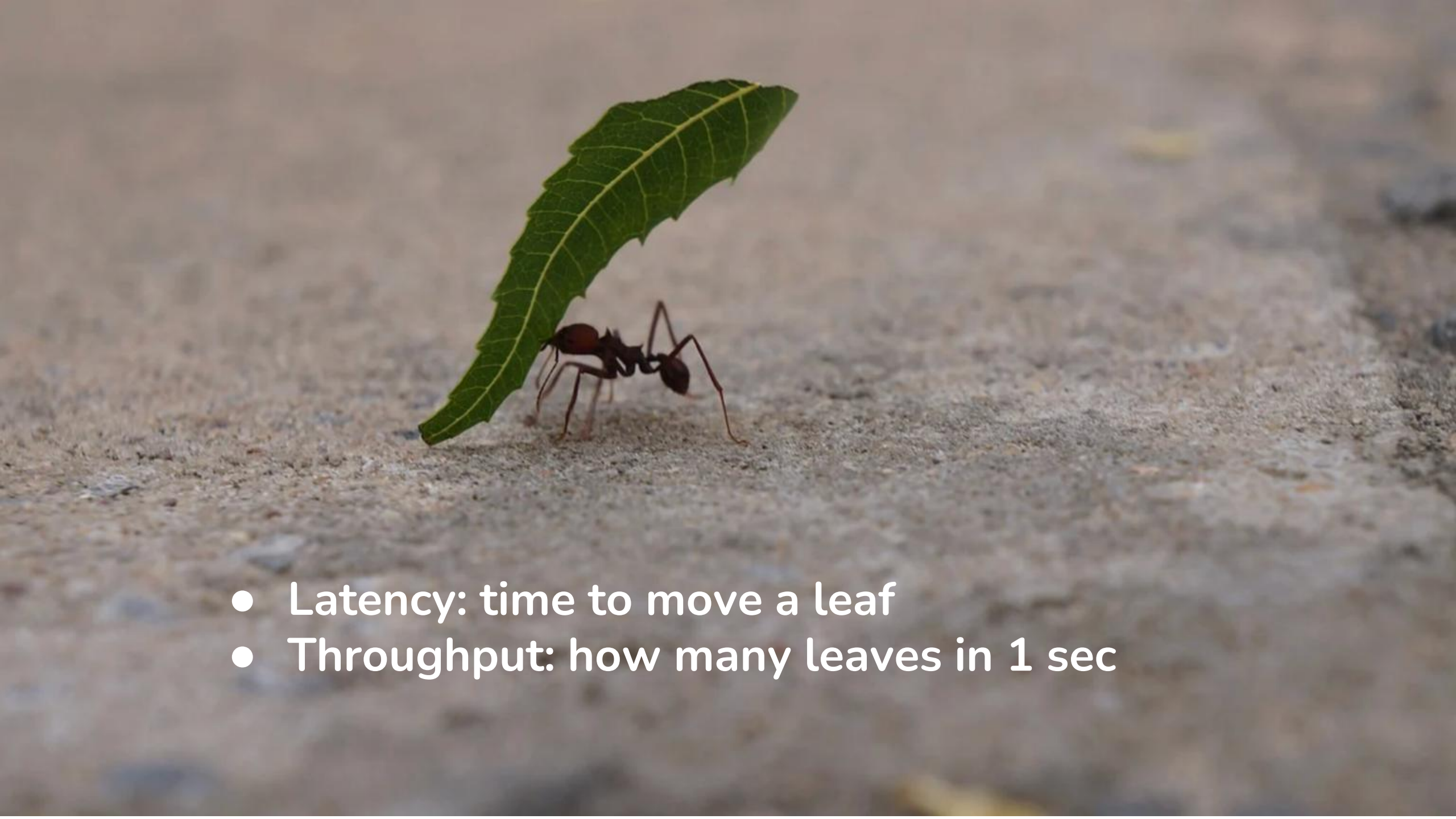
generating predictions

ประสิทธิภาพด้านการคำนวณ
จากผู้ใช้จริง

Latency matters

- 100ms delay can hurt conversion rates by 7% (Akamai study ‘17)
- 30% increase in latency costs 0.5% conversion rate (Booking.com ‘19)
- 53% phone users will leave a page that takes >3s to load (Google ‘16)

ข้อจำกัดด้านเวลาที่ตอบสนองจากระบบ



- Latency: time to move a leaf
- Throughput: how many leaves in 1 sec

- Real-time: low latency = high throughput
- Batched: high latency, high throughput

งานจริงต้องเป็น Real-time ไม่ใช่ Batched

ML in research vs. in production

	Research	**Production**
Objectives	Model performance	Different stakeholders have different objectives
Computational priority	Fast training, high throughput	Fast inference, low latency
Data	Static	Constantly shifting

เทียบด้านข้อมูลที่ใช้
shifting? หมายถึงอะไร

Data

Research	Production
● Clean ● Static ● Mostly historical data	● Messy ● Constantly shifting ● Historical + streaming data ● Biased, and you don’t know how biased ● Privacy + regulatory concerns

THE COGNITIVE CODER

By Armand Ruiz, Contributor, InfoWorld | SEP 26, 2017 7:22 AM PDT

The 80/20 data science dilemma

Most data scientists spend only 20 percent of their time on actual data analysis and 80 percent of their time finding, cleaning, and reorganizing huge amounts of data, which is an inefficient data strategy

การจัดการข้อมูลจำนวนมาก เสียเวลาไป
กับการจัดหาเตรียม data ให้เหมาะสม
ก่อนใช้จริง 80%

ML in research vs. in production

	Research	**Production**
Objectives	Model performance	Different stakeholders have different objectives
Computational priority	Fast training, high throughput	Fast inference, low latency
Data	Static	Constantly shifting
Fairness	Good to have (sadly)	Important

เทียบด้านความเป็นธรรม ไม่กีดกัน

Fairness

Fairness ด้าน Gender ไม่มี racism
คนจน คนรวย ... etc.

The Berkeley study found that both face-to-face and online lenders rejected a total of 1.3 million creditworthy black and Latino applicants between 2008 and 2015. Researchers said they believe the applicants "would have been accepted had the applicant not been in these minority groups." That's because when they used the income and credit scores of the rejected applications but deleted the race identifiers, the mortgage application was accepted.

ML in research vs. in production

ML Dev in Research ยังคงเป็น Model-Centric View หมายถึงเน้นการสร้าง learned model ที่มีประสิทธิภาพสูงสุด ยังไม่ดูแลส่วนอื่น ของระบบจริงที่ควรมีใน SE เรียกว่า System-Wide View

	Research	**Production**
Objectives	Model performance	Different stakeholders have different objectives
Computational priority	Fast training, high throughput	Fast inference, low latency
Data	Static	Constantly shifting
Fairness	Good to have (sadly)	Important
Interpretability	Good to have	Important

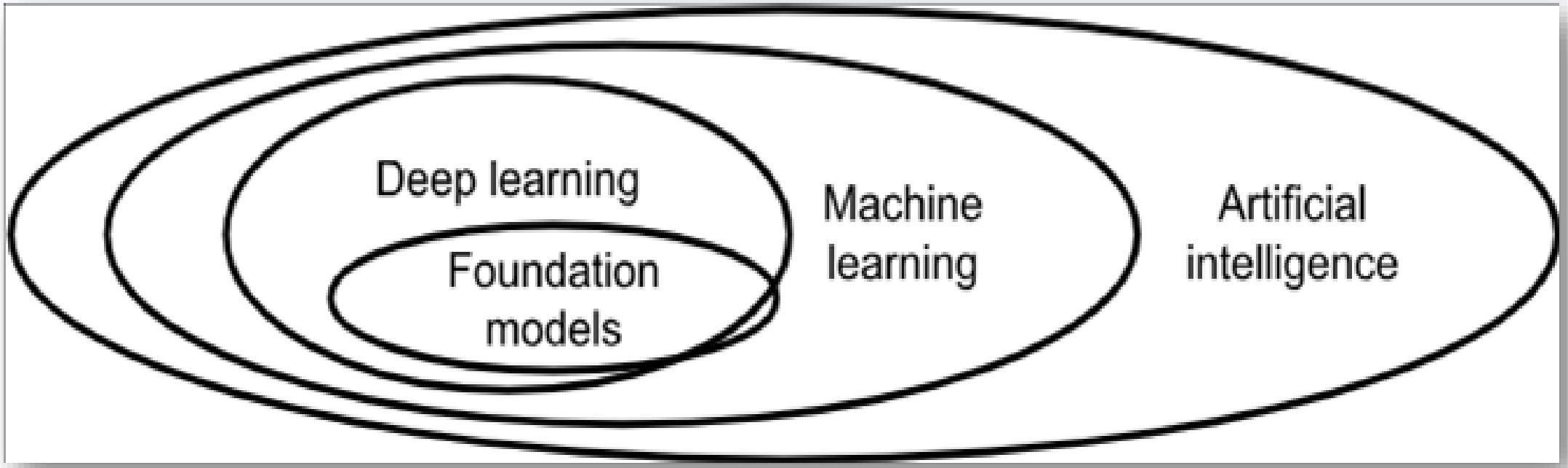
รู้ว่าผลลัพธ์เกิดจากการคำนวณอย่างไร มีที่มาที่ไป อธิบายได้

Intro to ML

มารู้จัก ML ให้มากกว่านี้

AI vs ML

เรามักไม่ใช้คำว่า AI ในปัจจุบันแล้ว
แต่เราจะใช้คำว่า ML แทน
เพราะ ML won AI war



AI is

- Artificial intelligence (AI) involves the fields of computer and data science focused on building machines with human intelligence to perform tasks like learning, reasoning, problem-solving, perception, and language understanding.

การเลียนแบบความฉลาดของมนุษย์
- อายุตนะทั้ง 6 : คือ ทาง ตา หู จมูก ปาก ลิ้น กาย ใจ
- ทำได้เกือบหมด ยกเว้น ใจ

AI : ML : DL

เปรียบเทียบคำที่ใช้

AI

- Non-learning AI
- Learning AI (become main approach of today's AI)

Non-learning AI

- If-the rule ie: chess game
- Expert system using Facts and Inference rules ie: medical diagnostic tool ดูจากอาการแล้ว วิเคราะห์หาโรค การรักษาจาก (fact and inference rule)
- Fuzzy logic AI
- Evolutionary Computation AI เช่น Genetic Algo.
- Classical Search & Planning ie: GPS computes the best path in graph

Learning AI (Machine Learning)

- Supervised Learning (data with labels) (classification, regression) มีครูสอน
- Unsupervised Learning (data no labels) (clustering, association) เรียนเอง
- Reinforcement learning (learn from environment with reward, penaly) (robot) ทำเอง แต่มีรางวัล
- learning from massive amounts of data

ML vs. Non ML

ML จะเรียนรู้จาก Data

Feature	Machine Learning (ML)	Non-ML AI (Symbolic/Logic)
Primary Driver	Data & Statistics	Human-written rules & Logic
Adaptability	Learns and improves from new data	Requires manual updates to rules
Transparency	Often a "black box" (hard to explain why)	Highly transparent (traceable logic)
Best For	Pattern recognition (Faces, Speech)	Strict logic (Legal, Accounting, Math)

ML → SL, UL, RL

การเรียนรู้จาก Data มีหลาย approaches

ประเภท ML	สิ่งที่ป้อนให้เป็น Input
Supervised	รูปส้ม 1,000 รูป + พร้อมบอกว่า "นี่คือส้มนะ"
Unsupervised	รูปผลไม้กองรวมกัน 1,000 รูป (ไม่บอกชื่อ) ให้ AI ไปหาความต่างเอง
Reinforcement	**Input:** คือตัวผลไม้ที่วางอยู่ตรงหน้า AI **Process:** AI ลองทายดูว่าคืออะไร **Feedback:** ถ้าทายถูก เราส่ง "รางวัล" (+1) ให้เป็น Input กลับไป

ML is unPredictable

- ML is incredibly powerful, but it requires **massive amounts of data** and can be unpredictable.
- If you are building a system for a nuclear power plant or a flight control system, you often want a **rule-based system** because you need to know exactly why a decision was made every single time

การใช้ ML มักคาดการณ์ไม่ได้
แต่ใช้ AI แบบอื่นจะคาดการณ์ได้

How ML works

ML แก้ปัญหาอะไรได้

Generic ML Problems

- Classification problem
- Regression problem

Classification problem

- binary classification
- multiclass problem

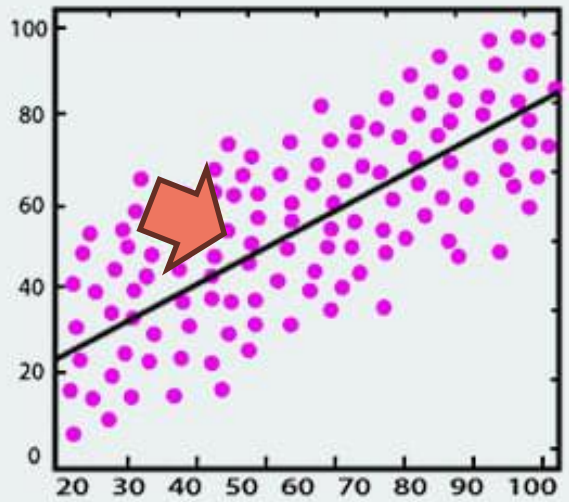
Multilabel problem…

- 1st solution - using encoding vector [1,0,1]
- 2nd solution - using separate model to each label

รูปตัวอย่าง Classification vs. Regression

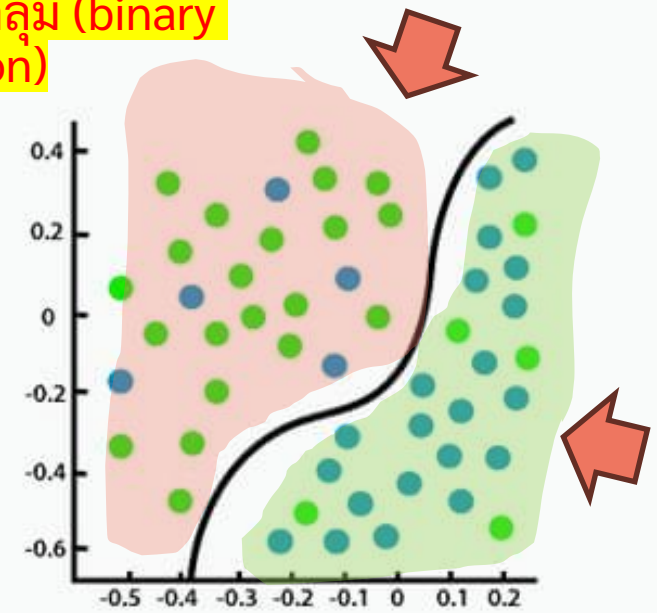
จากจุด data จำนวนมาก
หา model ตัวแทนเพื่อ
predict unseen data เช่น
เส้นตรงนี้ และ predict จุดอื่น
ๆ จาก model

จากจุด data จำนวนมาก
หา model ตัวแทนที่แบ่งกลุ่ม
ของจุด เช่น เส้นกราฟนี้เพื่อ ใช้
แบ่งกลุ่ม 2 กลุ่ม (binary
classification)



Regression

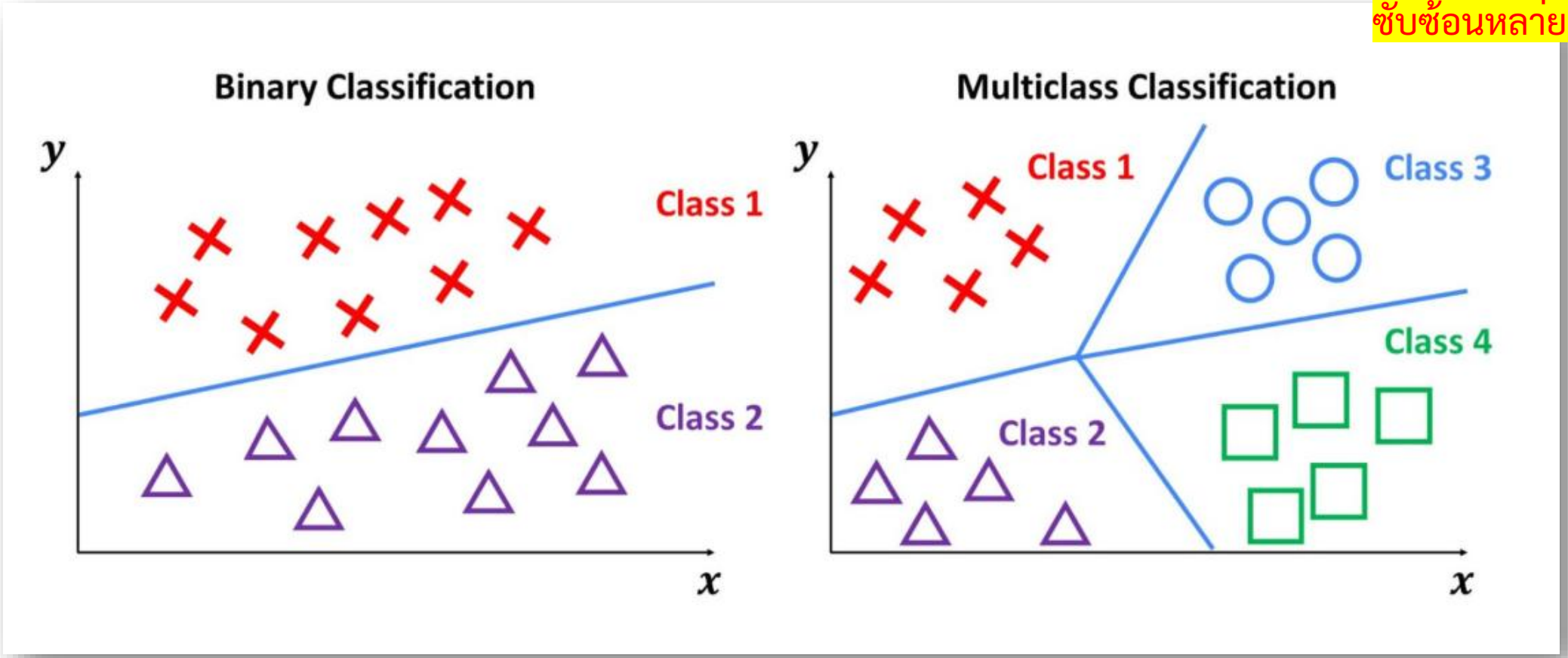
versus



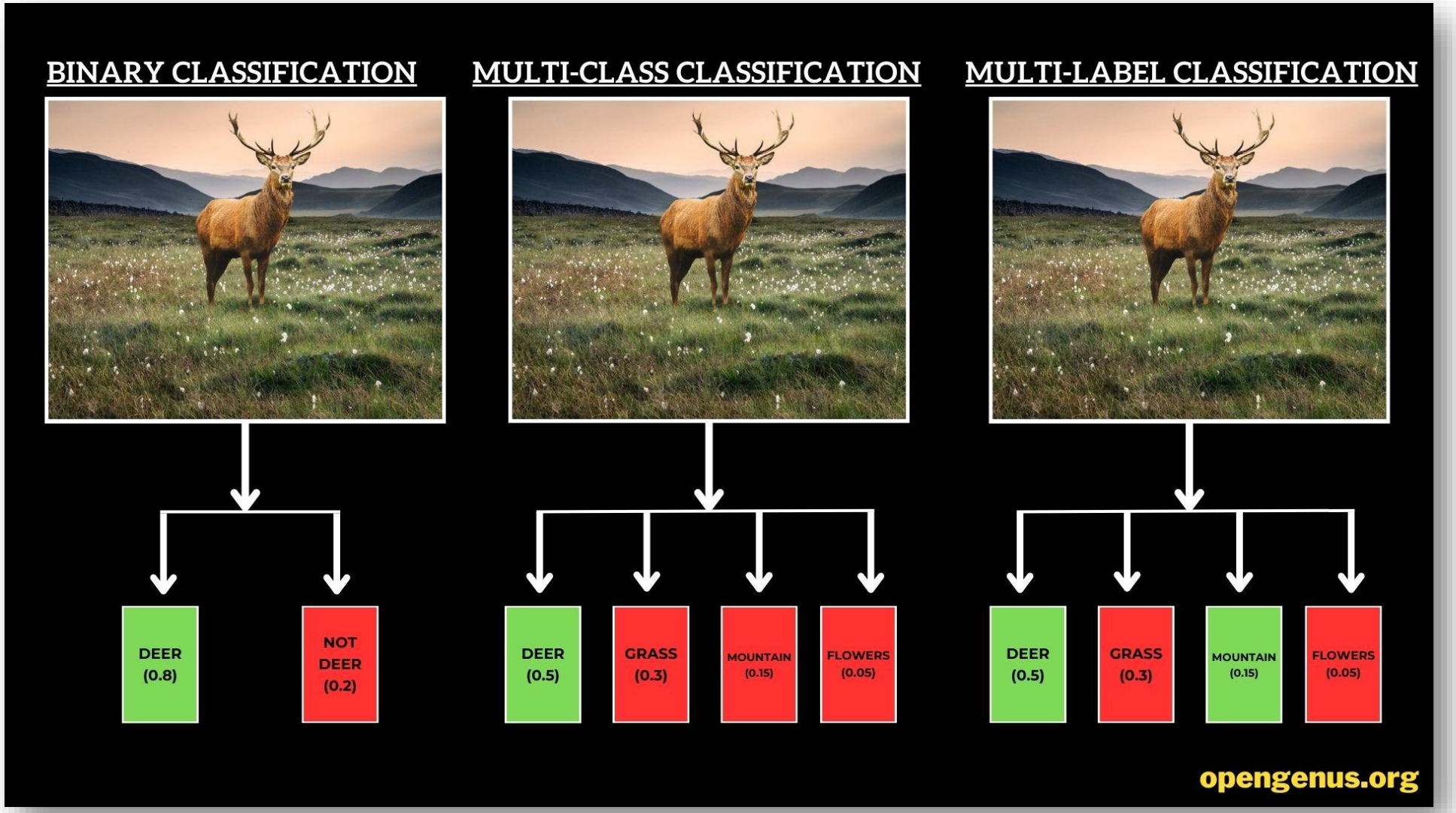
Classification

รูปตัวอย่าง Binary classification vs. Multiclass classification

Multiclass problem
ซับซ้อนหลายมิติได้



รูปตัวอย่าง Multi-label classification



รูปตัวอย่าง Multi-label classification

Multi-label classification (MLC) มักจะ encode output ให้เป็น vector หรือ matrix ที่มี 1 พร้อมกันได้หลายตำแหน่ง

Single-Label Classification

Data Case	Cat	Dog
	1	0
	1	0
	0	1
	1	0

Multi-Label Classification

Data Case	Cat	Grass	Tree	Cow	Rainbow
	0	1	0	0	1
	0	1	0	1	0
	1	0	1	0	0
	1	1	0	1	0

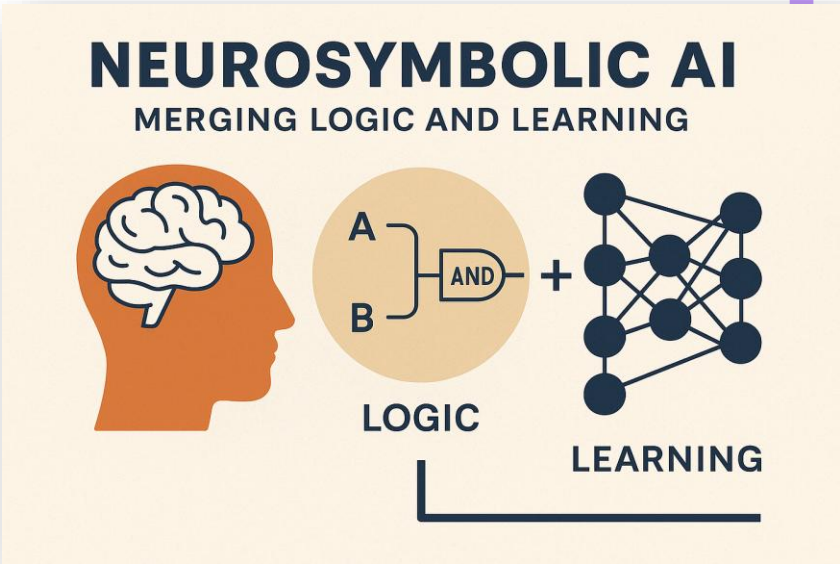
The Next Big Thing: Neuro-Symbolic AI

The biggest trend right now isn't choosing between "Learning" and "Rules"—it's combining them. This is called **Neuro-Symbolic AI**.

> **The Goal:** Combine the **perception** of Machine Learning (seeing and hearing) with the **reasoning** of Symbolic AI (logic and rules).

- **The ML part:** Identifies that there is a "stop sign" in a blurry photo.
- **The Symbolic/Logic part:** Understands the "Rule" that you must stop at that sign, regardless of what the data says

AI Trend → ML+Symbolic
เพื่อลด unpredictable



The Next Big Thing: Neuro-Symbolic AI

Approach	Popularity	Main Use Case	Reliability
Machine Learning	**High** (Dominant)	Generative AI (ChatGPT), Vision, Ads	High for patterns; low for facts
Rule-Based AI	**Medium** (Specialized)	Tax software, Medical logic, Calculators	100% predictable; zero flexibility
Hybrid (Neuro-Sym)	**Rising** (Future)	Robotics, Scientific Research, Legal AI	Aims for the best of both worlds

ML vs. Deep Learning

To understand the difference, it’s best to see them as a "nesting doll" relationship. **Deep Learning (DL)** is not a competitor to **Machine Learning (ML)**; it is a specialized, more complex version of it

The Key Difference: Feature Extraction

The biggest practical difference is how the computer "sees" the data.

- **Standard ML:** A human expert usually has to tell the computer what to look for. If you want to identify a "car," you might manually program features like *wheels, windshield, and headlights*.
- **Deep Learning:** You just give it millions of photos of cars. The "Deep" neural network figures out the features (wheels, shapes, textures) on its own by passing the data through many layers of artificial neurons

ML and Deep Learning

Feature	Machine Learning (Traditional)	Deep Learning (Advanced)
Data Size	Works well with small/medium data.	Needs **massive** amounts of data.
Hardware	Can run on a standard laptop CPU.	Requires high-end **GPUs** or TPUs.
Preparation	Needs a human to "clean" and pick features.	"Feature extraction" is automatic.
Logic	Easier to explain (Transparent).	Very hard to explain (Black Box).
Training Time	Seconds to hours.	Days to weeks.

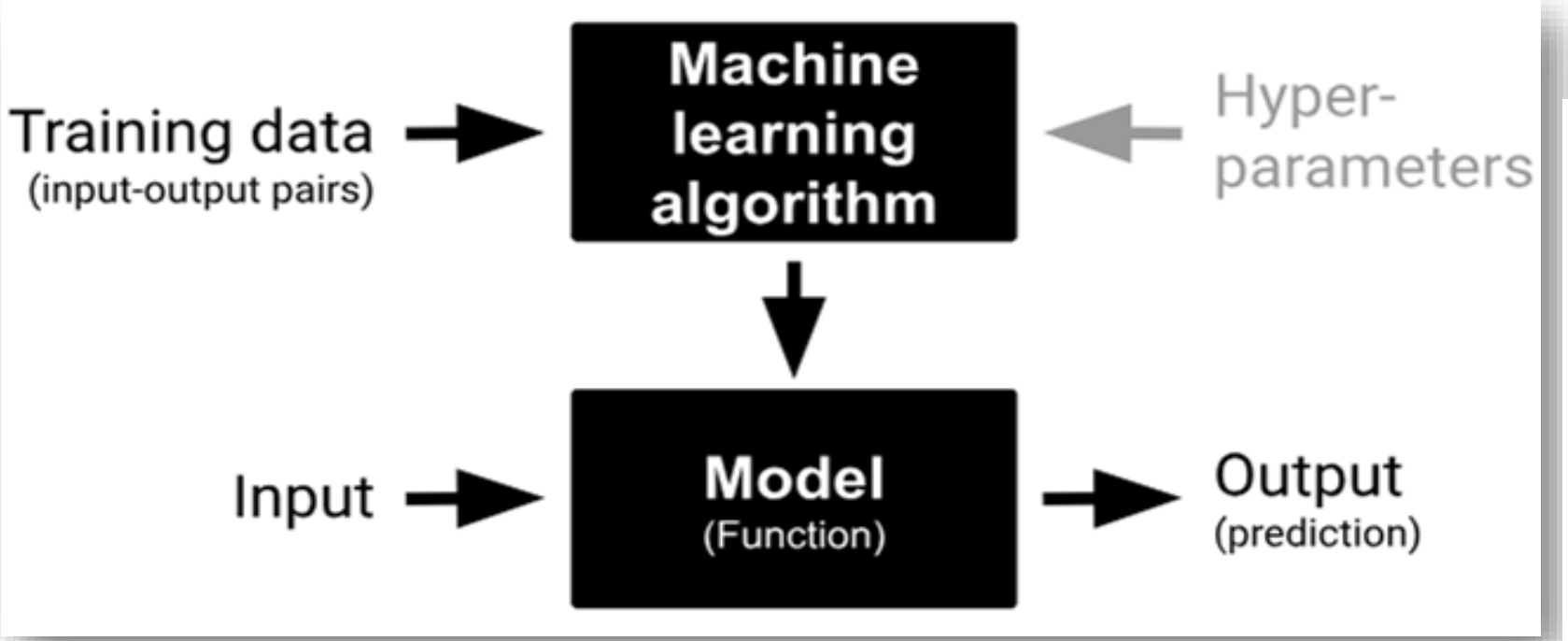
คำศัพท์ที่ใช้ในวงการ

ML Algorithm (Modelling Technique) (1)

- A machine-learning algorithm is implemented in a machine-learning library or machine-learning framework, such as sklearn or Tensorflow
- It defines the training procedure of how the function is learned from the observations.
- The learned function is called machine-learned model
- The action of feeding observations into a machine-learning algorithm to create a model is called model training.

ML Algorithm (Modelling Technique) (2)

- Supervised ML algorithms learn from observations in the form of pairs of data and corresponding label, where the label describes the expected output for that data.
- The learned model computes outputs for inputs from the same domain as the training data—for example, a model trained to generate captions for images takes an image and returns the caption.
- These outputs are often called predictions, and the process of computing a prediction for an input is called model inference.



- ถ้าเทียบเคียงกับ การเขียนโปรแกรมแล้ว
- Training data คือ Source code
- ML algo. เหมือน compiler ที่สร้าง model function)
- ตอนใช้งาน ML algo. ไม่เกี่ยวแล้ว แต่จะใช้ model แทน

input → executable code (program byte code) → output

source code is compiled by compiler

input → machine learned model (model function) → output

training data (=source code) is trained by ML algo (=compiler) and hyperparameters (=compiler options)

Model Architecture

- The process of training a model is often computationally intensive
- Models to be learned typically follow a certain basic internal structure specific to that learning algorithm, such as if-then-else chains in decision trees and sequences of matrix multiplications in deep neural networks.
- In deep learning, this internal structure is called model architecture.
- During training, the machine-learning algorithm then identifies the values of constants and thresholds within the internal structure, such as learning the conditions of the if-then-else statements or the values of the matrices—those constants and thresholds are called model parameters.

Model Parameters

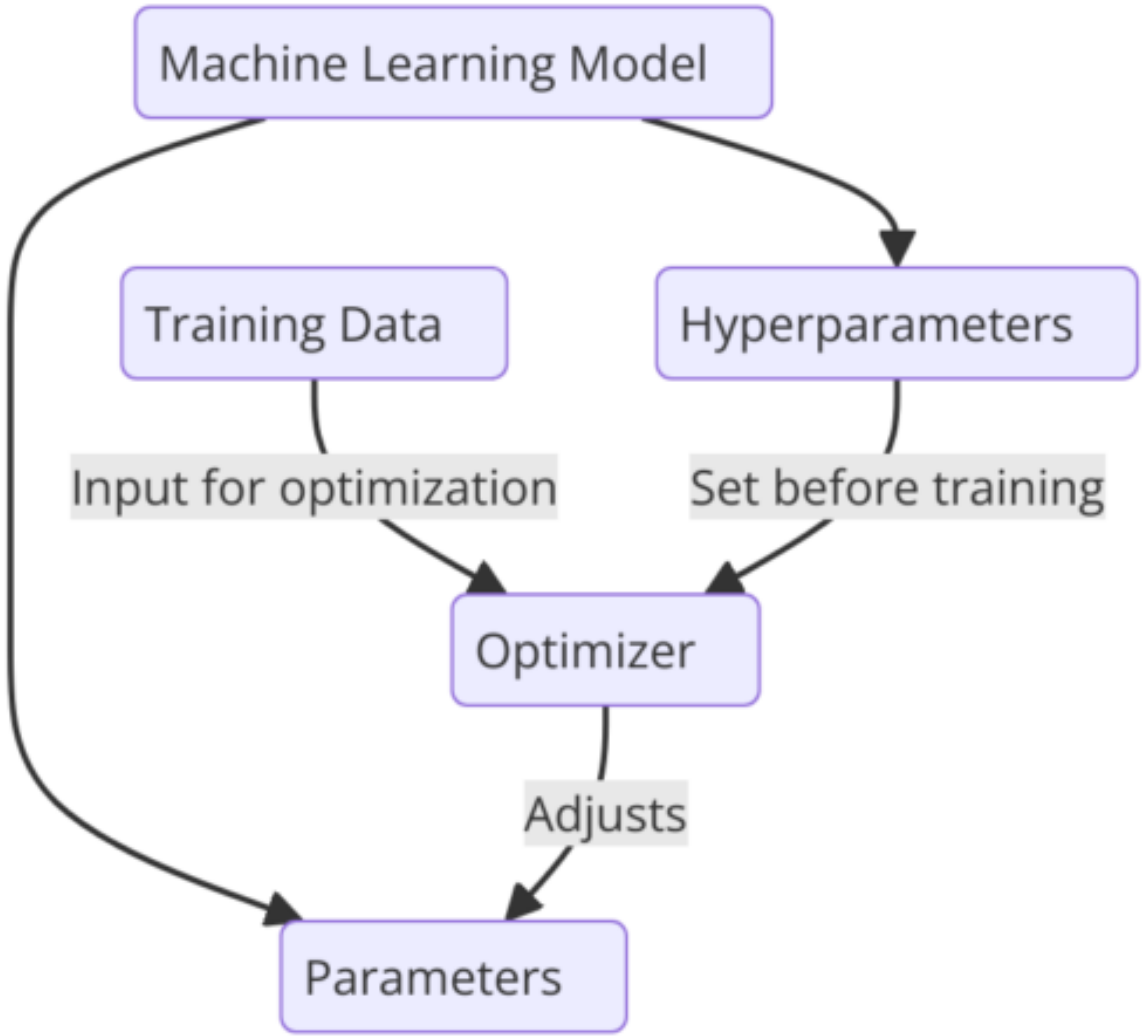
- Model parameters are internal variables of the model that are estimated or learned automatically from the data during the training process.
- They define the specific characteristics of the model after it has been trained.
  - **Learned from data**: The algorithm determines the best values for these parameters as it processes the training data
  - **Internal to the model**: They are part of the final, trained model
  - **Examples**:
    - The weights and bias in neural network
    - The coefficients in linear or logistic regression

Hyperparameters

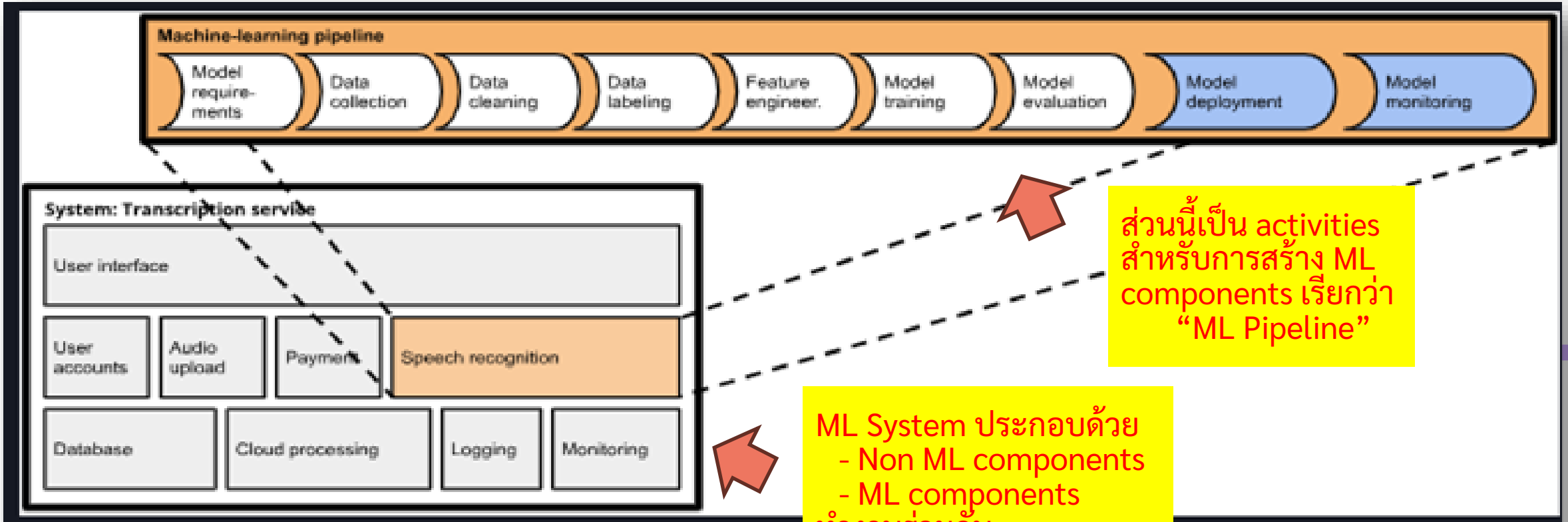
- Hyperparameters are external configurations or settings that are set manually by the practitioner before the training process begins.
- They control how the learning process itself is performed
- The model does not learn these values from the data
  - **Set manually**: The data scientist must choose these values based on experience or by using techniques like grid search or random search
  - **External to the model**: They are part of the training algorithm, not the final model itself
  - **Examples**:
    - *Learning rate*: Controls how much the model changes in response to estimated error each time the model weights are updated
    - *Number of epochs*: How many times the entire training dataset is passed through the learning algorithm
    - *Batch size*: The number of samples used in one iteration of training

Model Parameters vs. Hyperparameters

Aspect	Parameters	Hyperparameters
Definition	Internal model values learned from data.	External configurations set before training.
Optimization	Optimized during training by the algorithm.	Not optimized, manually set or tuned.
Examples	Weights, biases in neural networks.	Learning rate, batch size, number of epochs.
Adjustment Process	Automatically adjusted by learning algorithms.	Manually set or adjusted through tuning.


Machine Learning Model
Training Data
Hyperparameters
Input for optimization
Set before training
Optimizer
Adjusts
Parameters

Machine Learning Pipeline (diagram)



ส่วนนี้เป็น activities สำหรับการสร้าง ML components เรียกว่า "ML Pipeline"

ML System ประกอบด้วย
- Non ML components
- ML components

ทำงานร่วมกัน

The ML pipeline corresponds to all activities for producing, deploying, and updating the ML component. The resulting ML component is part of a larger system.

Machine Learning Pipeline (1)

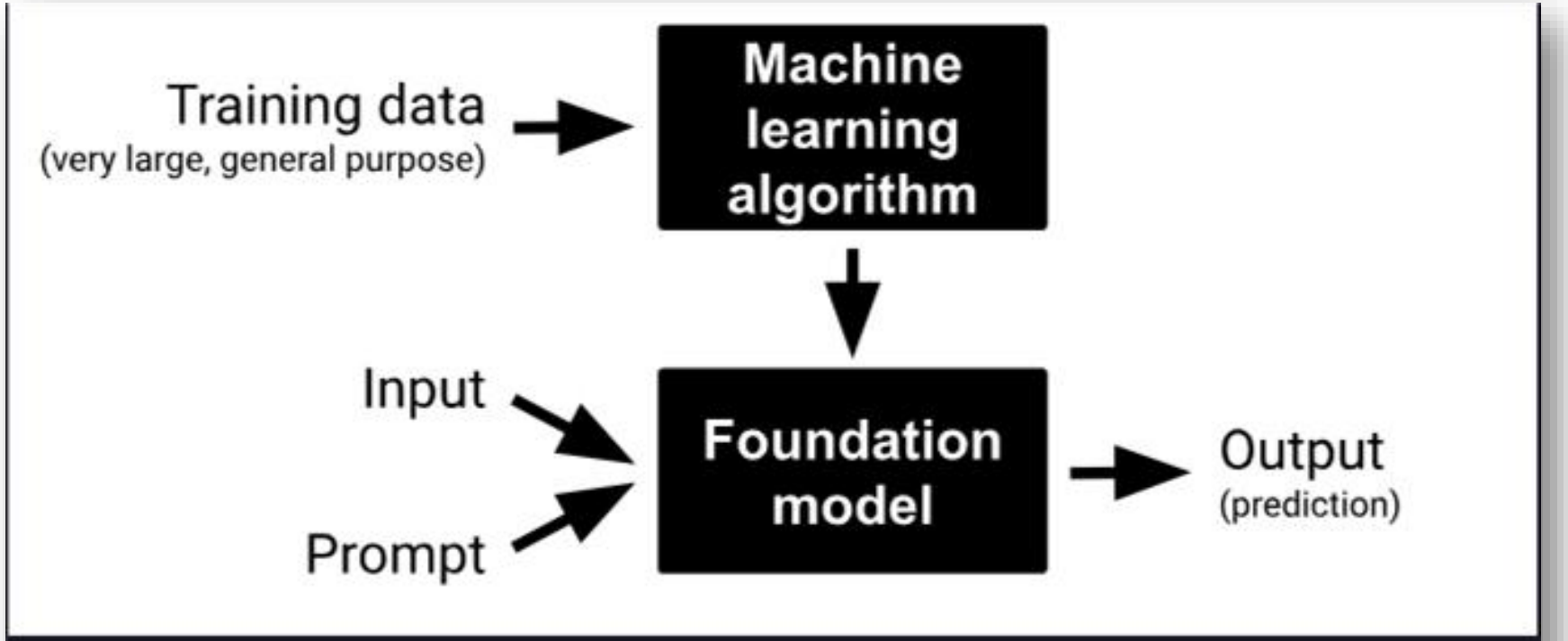
- Once the purpose or goal of the model is clear (model requirements).
- But before the model can be trained, we have to acquire training data (data collection), identify the expected outcomes for that training data (data labeling), and prepare the data for training (data cleaning and feature engineering).
- The preparation often includes steps to identify and correct mistakes in the data, fill in missing data, and generally convert data into a format that the machine-learning algorithms can handle well.
- After the model has been trained (model training) with a machine-learning algorithm, it is typically evaluated (model evaluation).
- If the model is deemed good enough, it can be deployed (model deployment), and may be monitored in production (model monitoring).

Machine Learning Pipeline Automation

- The entire process of developing models is highly iterative, incrementally tweaking different parts of the pipeline toward better models.
- For example, if the model evaluation is not satisfactory, data scientists might try different machine-learning algorithms or different hyperparameters, might try to collect more data, or might prepare data in different ways.
- Most steps of the machine-learning pipeline are implemented with some code. For example, data preparation is often performed with programmed transformations rather than manual changes to individual values
- Automation tools are needed
  - Computational notebooks like Jupyter are common for exploratory development with code cells for data cleaning, feature engineering, training, and evaluation. The entire process from receiving raw data to deploying and monitoring a model can be automated with codef

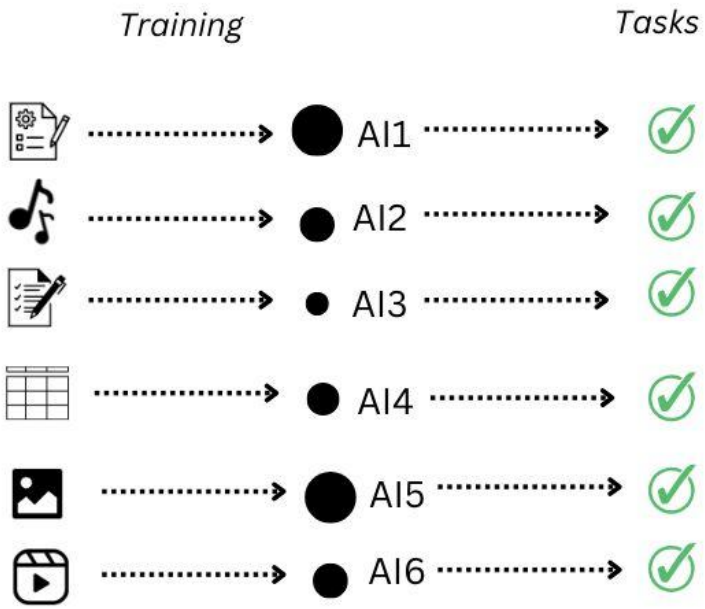
Foundation Models and Prompting

- The recent rise of large language models like GPT-4 has triggered a shift in how some projects approach machine learning.
- Rather than learning a model for each task, organizations train very large general-purpose models, called foundation models as an umbrella term for large language models and other large general-purpose models.
- Those foundation models can be instructed to perform specific tasks with prompts.
- For example, instead of developing a toxicity detection model trained on examples of toxic and non-toxic language, we can send a prompt like “Answer only yes or no.
  - Is the following sentence toxic: [input]” to a foundation model and expect it to answer without any specific training for toxicity.



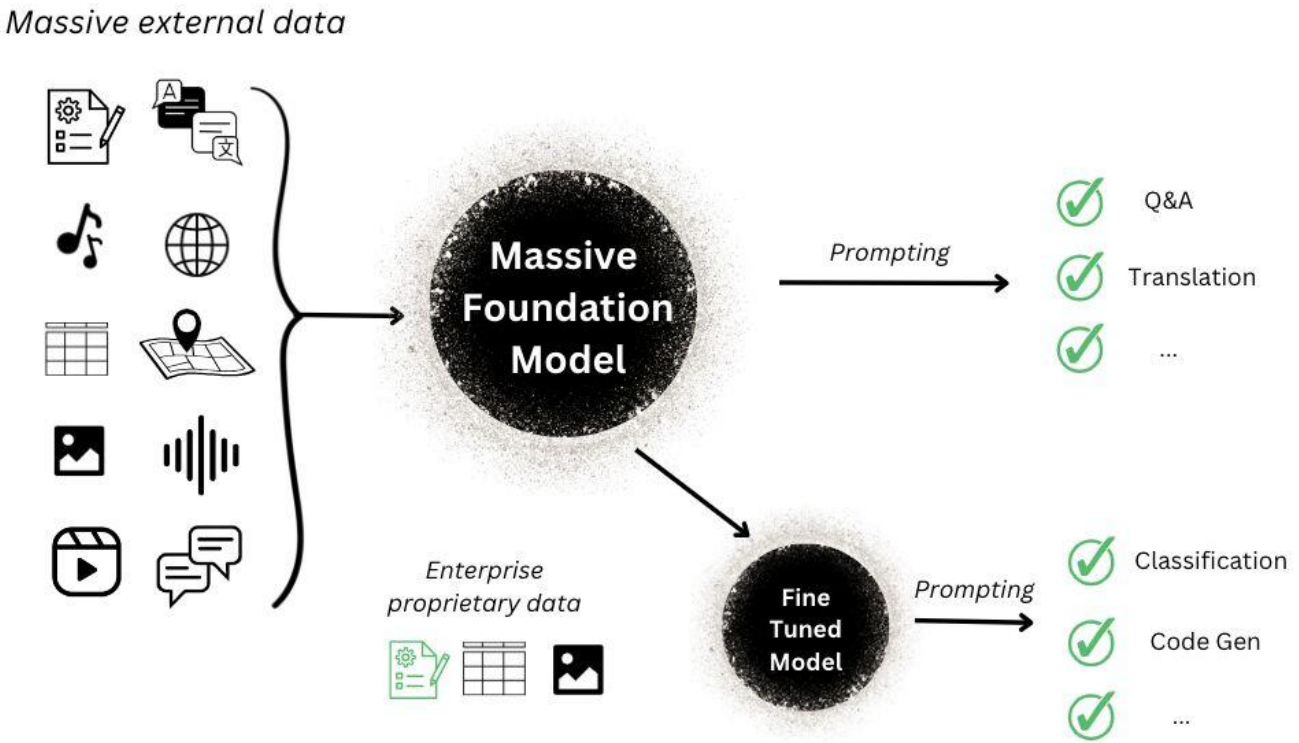
Foundation models are general-purpose models created with machine-learning algorithms and a prompt customizes the model for a given input.

Traditional ML



- Individual siloed models
- Require task-specific training
- Lots of human supervised training

Foundation Models



- Massive multi-tasking model
- Adaptable with little or no training
- Pre-trained unsupervised learning

Sample of the Existing Foundation Models

- Foundation models are large-scale artificial intelligence models trained on vast, diverse datasets—often using self-supervised learning—that can be adapted to a wide range of downstream tasks.

DINOv2 (DIstillation with NO labels v2) is an open-source **self-supervised computer vision model** developed by Meta AI. It is designed to learn high-quality, general-purpose visual features from massive amounts of unlabeled data, functioning as a "foundation model" for vision tasks

Whisper is an automatic speech recognition (ASR) system trained on 680,000 hours of multilingual and multitask supervised data collected from the web. We show that the use of such a large and diverse dataset leads to improved robustness to accents, background noise and technical language. Moreover, it enables transcription in multiple languages, as well as translation from those languages into English. We are open-sourcing models and inference code to serve as a foundation for building useful applications and for further research on robust speech processing.

ML System Design

How difficult?

Why ML Systems Design?

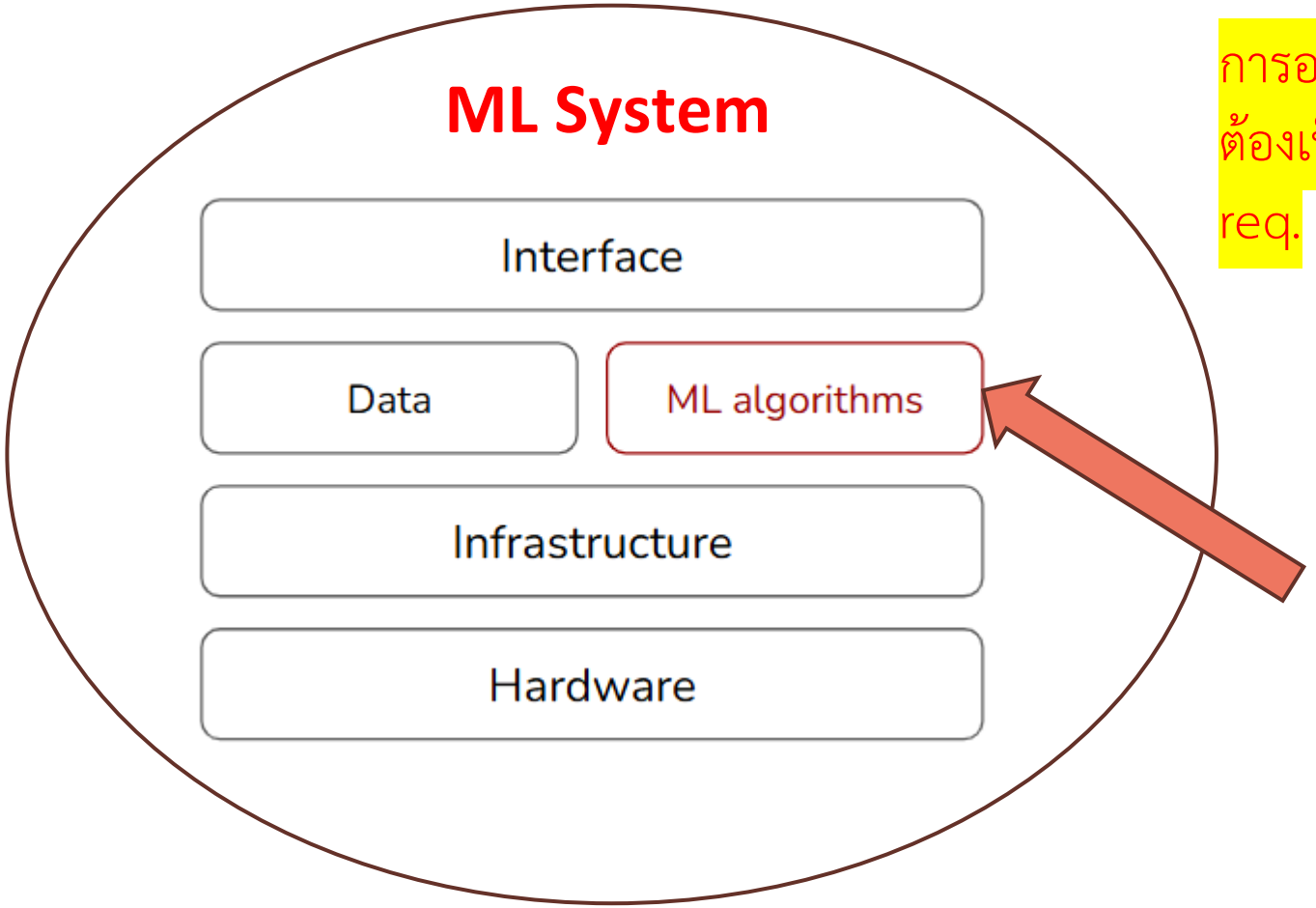
- ML algorithms is the less problematic part.
- The hard part is to **how to make algorithms work with other parts to solve real-world problems**.
- 60/96 failures caused by non-ML components

ทำอย่างไรให้ ML algo. ทำงานร่วมกับส่วนอื่นของระบบซอฟต์แวร์ได้จริงจัง

ML Dev in Research ยังคงเป็น "Model-Centric View" หมายถึงเน้นการสร้าง learned model ที่มีประสิทธิภาพสูงสุด ยังไม่ดูแลส่วนอื่น ของระบบจริงที่ควรมีใน SE เรียกว่า "System-Wide View"

Machine Learning Systems Design

- The process of defining the **interface, algorithms, data**, **infrastructure**, and **hardware** for a machine learning system to satisfy **specified requirements**. (reliable, scalable, maintainable, adaptable)
- The hard part is to how to make algorithms work with other parts to solve real-world problems



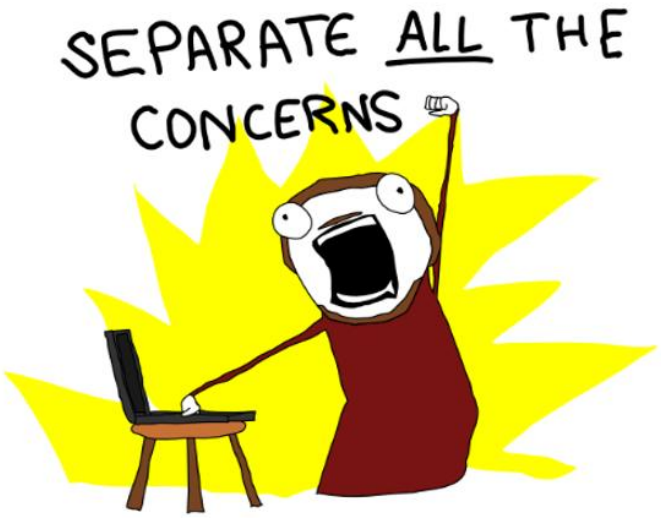
การออกแบบ ML Sys แรกเริ่มต้องเป็นไปตาม non-functional req.

ML algorithms are only this part

Traditional Software

- Separation of concerns is a design principle for separating a computer program into distinct sections such that each section addresses a separate concern
- Code and data are separate
  - Input data into the system should not change the underlying code

เพราะ traditional sw จำเป็นต้องเป็นไปตาม functional & non-functional req.



ML Systems (MLS)

MLS = Code + Data

- They consume a massive amount of data, require heavy computational power
- Code and data are tightly coupled
  - ML systems are part code, part data
- Not only test and version code, need to test and version data too

Machine learning is an approach to (1) learn (2) complex (3) patterns from (4) existing data and use these patterns to make (5) predictions on (6) unseen data.

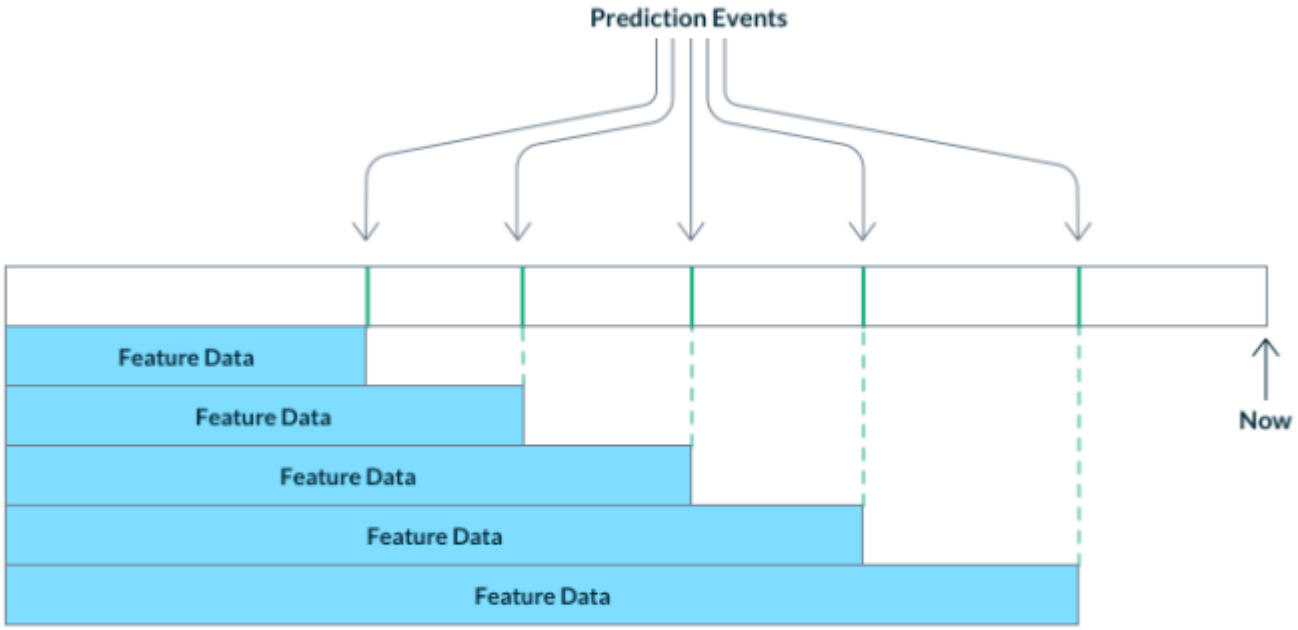
การจัดการ data (test and versioning) ยากกว่า เพราะอะไร?

Test and version data

- Extremely hard to ensure correctness in time

ทำไมการจัดการ data จึงยากกว่า?

- **Feature Data** (ข้อมูลคุณลักษณะ) ในรูปภาพ หมายถึง **ข้อมูลอินพุต** ที่ใช้ในการฝึกฝนและใช้งานโมเดล Machine Learning (ML)
- เมื่อเวลาผ่านไป Feature data จะแตกต่างกัน feature ที่เก็บจะมีการเปลี่ยนแปลงแบบกระจายตัว (เกิด data distribution) ทำให้ต้องมีการ retrain ML model ใหม่



Timestamp	Label	User ID	Feature Value
2:00	1	1	5
3:00	0	1	19
3:30	0	1	21
5:00	1	1	27
6:00	1	1	42
7:30	0	1	55

ML Systems: version data

- Line-by-line diffs like Git does not work with datasets
- Cannot naively create multiple copies of large datasets
- How to merge changes?

สำหรับ code เราทำ to version code using diffs

Version control systems use diffs to:
1.Track changes — know what lines changed, who changed them, and when.
2.Merge branches — combine code from multiple people safely.
3.Revert changes — go back to a previous version easily.
4.Optimize storage — store only changes (diffs), not whole copies of every file.
5.Review code — in pull requests, diffs help reviewers see exactly what was modified.

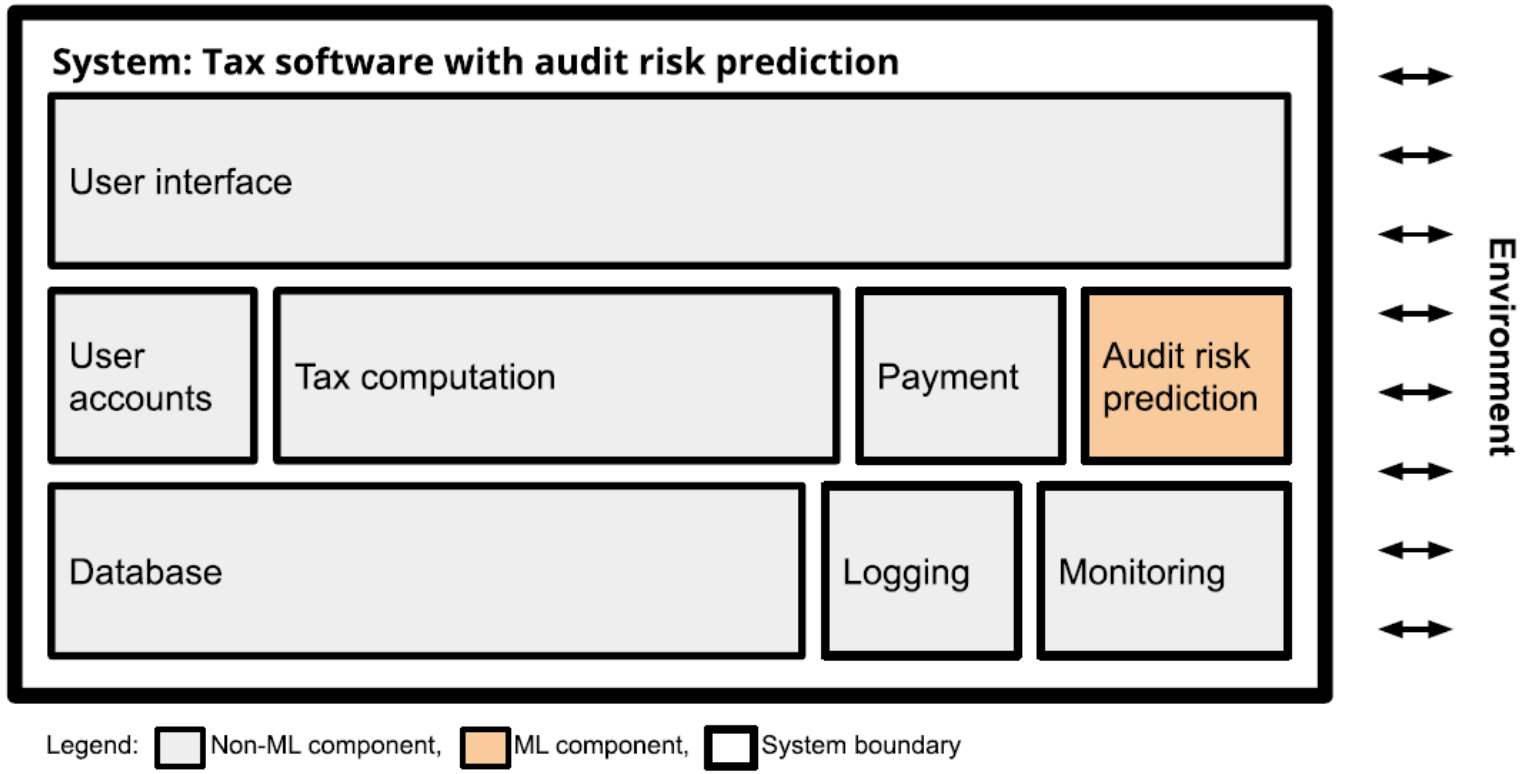
How to

"ถ้า Data Distribution ของวันนี้ ไม่เหมือนกับ Data Distribution ของวันแรกที่เทรนโมเดล... เราเรียกว่าเกิด Data Drift"

เมื่อเกิด Data Drift ผลลัพธ์คือ โมเดลจะเริ่มทายผิดมากขึ้นเรื่อยๆ โดยที่ไม่มี Error แจ้งเตือนในระบบ (Silent Failure)

- Validate data correctness?
- Test feature's usefulness?
- Detect when the underlying data distribution has changed?
- Know if the changes are bad for models without ground truth labels?
- Detect malicious data? (ข้อมูลประสงค์ร้าย ที่ถูกสร้างมาโจมตีโดยเจตนาทำผิด)
  - Not all data points are equal
  - Bad data might harm your model and/or make it susceptible to attacks

It is the process of ensuring that the *variables* (features) fed into your model are correct, consistent, and useful.

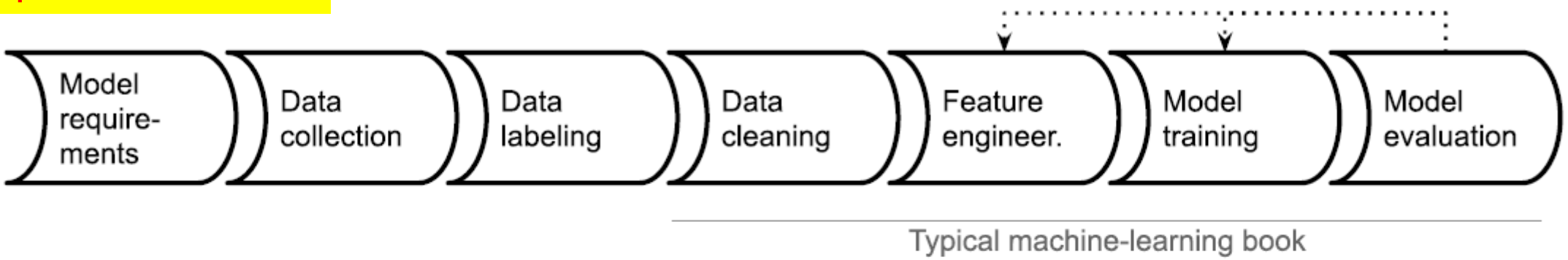


An architecture sketch of the tax system, illustrating the ML component for audit risk as an addition to many non-ML components in the system.

การออกแบบ MLS เราแค่มอง ML component เป็น function ที่เพิ่มมาให้ใช้งานเท่านั้นไม่ได้

Traditional Model Focus (Data Science)

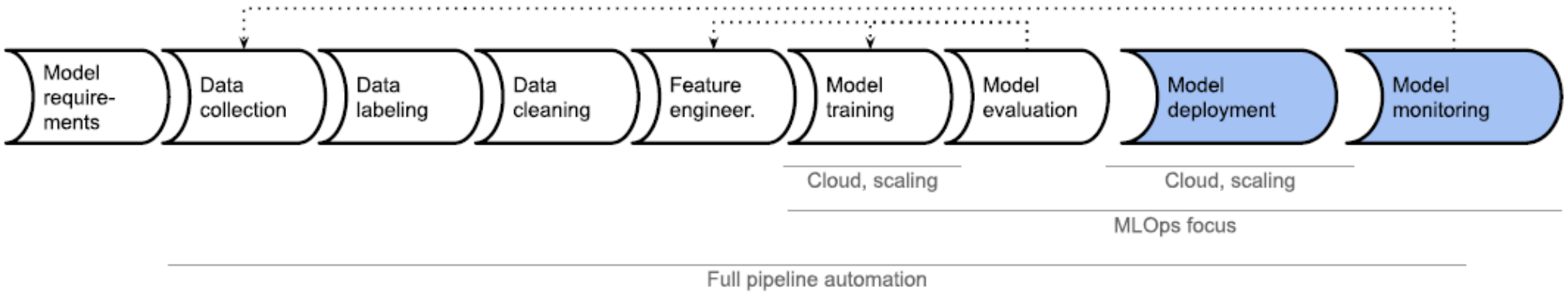
ML Pipeline แบบเดิมที่ใช้กัน
อยู่ในกลุ่ม data scientist



Typical steps of a machine-learning process. Mainstream machine-learning education and research focuses on the modeling steps itself with provided datasets.

Automating Entire ML Pipeline in Production Systems

ML Pipeline แบบใหม่ที่ใช้กันเพื่อระบบ MLS จริงจัง



Machine-learning practitioners working on production systems increasingly widen their focus from modeling to the entire ML pipeline, including deployment and monitoring, with a heavy focus on automation.

ในคาบถัด ๆ ไป
จะเป็นการลงรายละเอียด
หัวข้อต่าง ๆ

VDO
AI Trends in 2026

https://www.youtube.com/watch?v=zt0JA5rxdfM